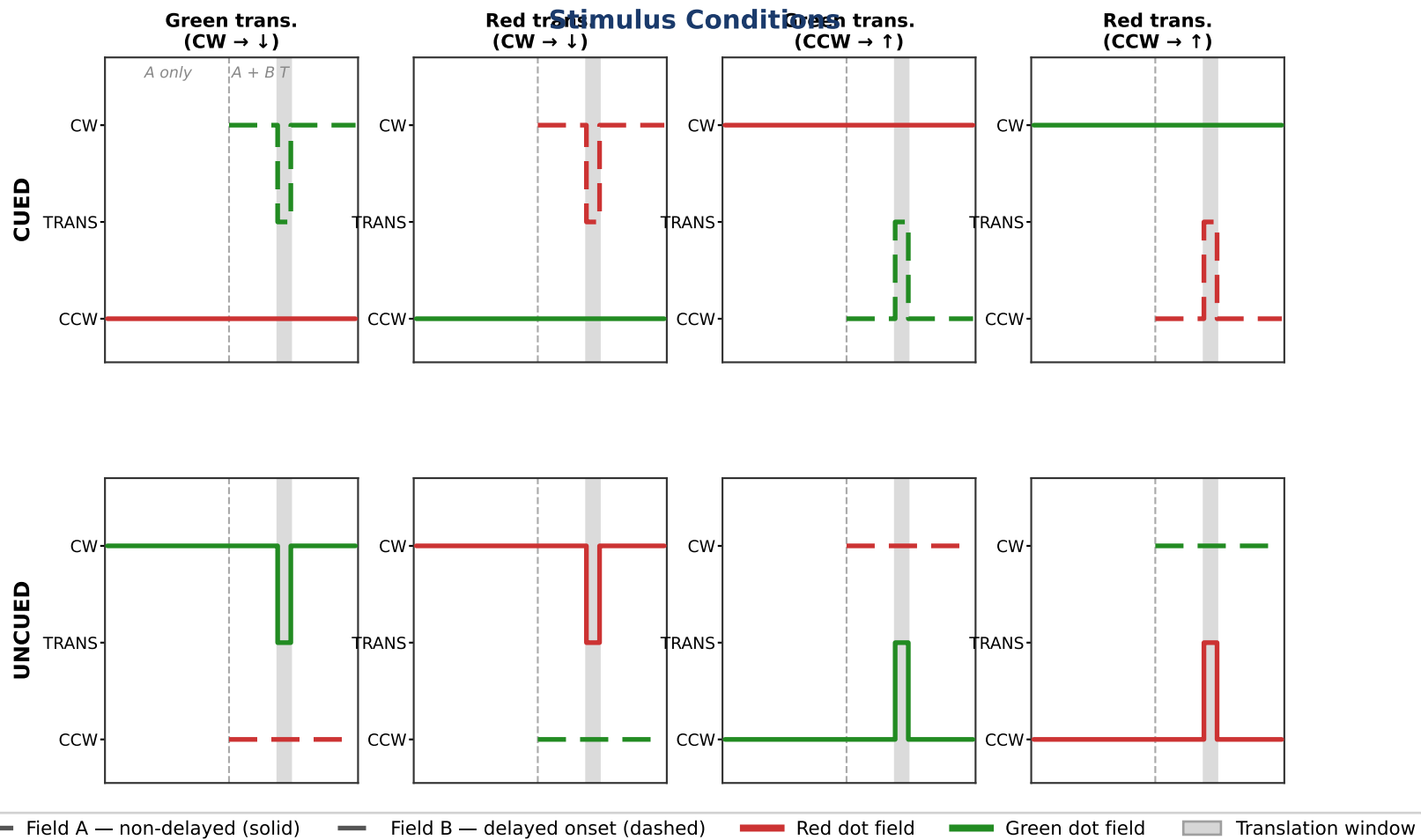


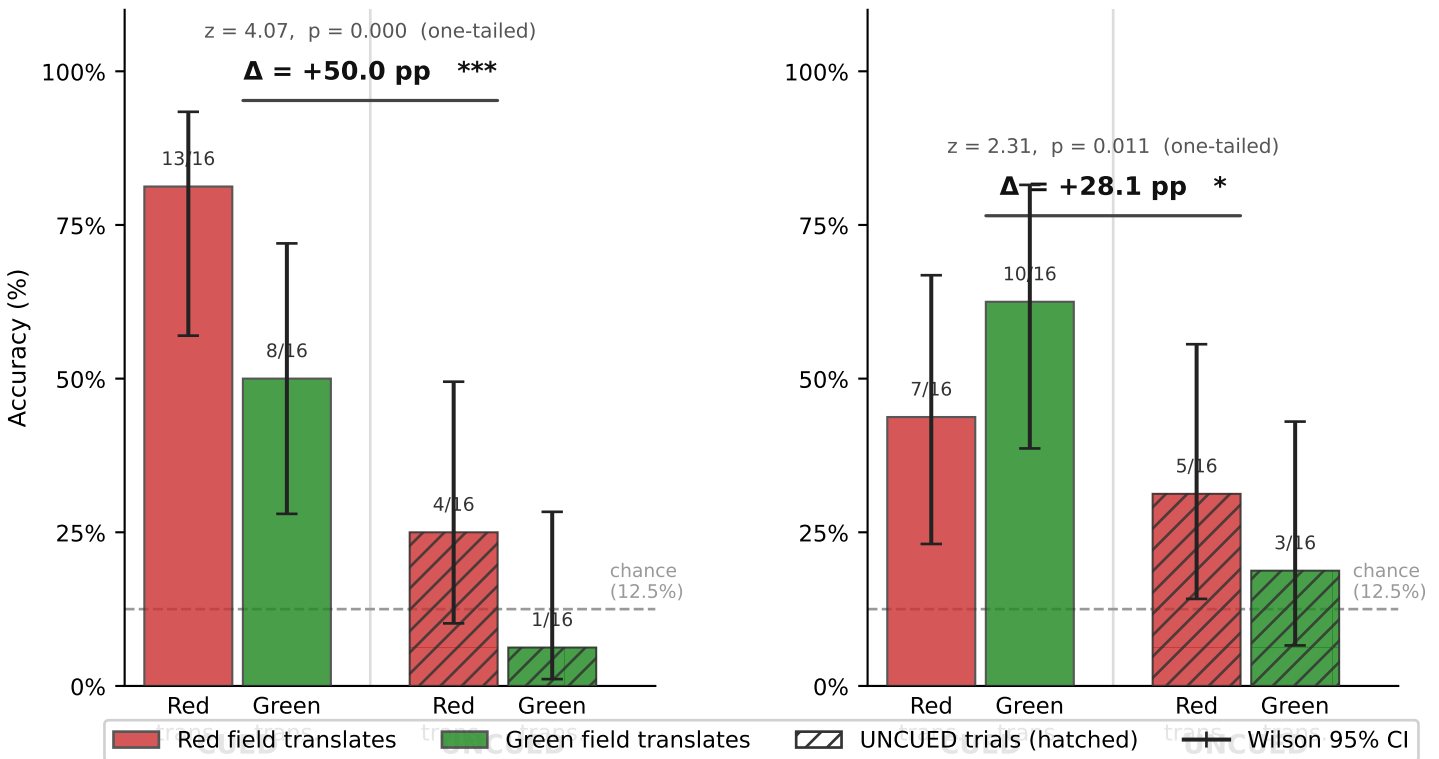
VRDots — Baseline Experiment: No Depth, No Swap

Sessions: 260323_1534 (pre-v0.2.0) 260324_0716 (v0.2.0)

Two baseline sessions were collected to establish the delayed-onset cueing effect in the Meta Quest 3 VR environment. Stimuli consisted of two superimposed, oppositely rotating dot fields (one red, one green; ~63 dots each, 81°/s, aperture radius 3.5°) presented on a flat fronto-parallel plane at 2.0 m with zero stereo depth separation. One field had a delayed onset of 750 ms (Field B, dashed lines), establishing a temporal onset asymmetry that serves as the implicit cue. After a 300 ms rotation period following Field B onset, one field translated briefly (80 ms, ~2.26°/s) in one of 8 equally spaced directions (0–315°, step 45°). On CUED trials the delayed-onset field (Field B) translated; on UNCUED trials the non-delayed field (Field A) translated. Subjects reported the translation direction via thumbstick (8-AFC). All 8 stimulus permutations shown in the figure above were presented (2 rotation configs × 2 delayed-field colors × CUED/UNCUED), balanced across 8 translation headings, giving 64 trials per session. No swap manipulations were applied in these sessions.



Each panel shows Field A (solid line) and Field B (dashed line) motion trajectories over trial time. The grey shaded region marks the 80 ms translation window. The vertical dashed line marks Field B onset. Row labels (CUED / UNCUED) indicate which field translates. Column labels indicate the translating field color and rotation direction. Line color indicates dot field color (red or green).

**Session 1 (260323_1534)
pre-v0.2.0****Results****Session 2 (260324_0716)
v0.2.0 [cursor-jump]****Findings**

Session 1 (260323_1534, pre-v0.2.0) yielded a strong cueing effect: CUED = 65.6%, UNCUEd = 15.6%, $\Delta = +50.0$ pp. A pronounced red-field advantage was present — accuracy was substantially higher when the red field translated than when the green field did, regardless of cueing status (CUED+Red = 81.2% vs. CUED+Green = 50.0%; UNCUEd+Red = 25.0% vs. UNCUEd+Green = 6.2%). This asymmetry most likely reflects imperfect iso-luminance equating: the heterochromatic flicker photometry calibration may not have fully equalised the perceptual salience of the two dot fields, leaving the red field physically more detectable. This session predates the v0.2.0 software version and should be treated as a preliminary observation.

Session 2 (260324_0716, v0.2.0) showed a reduced cueing effect: CUED = 53.1%, UNCUEd = 25.0%, $\Delta = +28.1$ pp. The red/green accuracy asymmetry was partially reversed relative to Session 1 (CUED+Red = 43.8%, CUED+Green = 62.5%), a pattern not yet explained — it may reflect sampling noise at $n = 16$ /cell or an interaction with the cursor-jump artifact described below. Despite these caveats, the cueing effect itself is robust and directionally consistent across both sessions, confirming that the delayed-onset manipulation produces reliable attentional cueing in the VR environment.

Software version and known artifacts

Software version v0.2.0 refers to the first formally version-stamped Quest build on the wip/quest-pilot branch. It introduced five changes relative to the pre-v0.2.0 build used in Session 1: (1) a subject-ID field and version number embedded in all data files, enabling post-hoc identification of which software generated a given dataset; (2) a 2-second delay before the response scene accepts thumbstick input, preventing the skip menu from auto-firing when the trigger remained held during a scene transition; (3) a corrected fixation-exclusion radius for dot spawning ($0.5^\circ \rightarrow 1.1^\circ$), matching the physical extent of the crosshair arms; (4) randomised trial order seeded with the system tick count and logged per session for reproducibility; and (5) a lowered thumbstick deadzone threshold ($0.5 \rightarrow 0.3$ normalised units), reducing the effort required to register a directional response.

Cursor-jump artifact (Session 2 only): the Quest thumbstick controller occasionally snapped the response cursor to an adjacent 45° direction sector without intentional user movement. Because each of the 8 AFC sectors spans 45° , a single snap produced a scored error even when the subject's intended direction was correct. The effect was identified post-hoc from the distribution of response errors, approximately 25% of which were adjacent-sector (off-by- 45°) responses — too frequent to be perceptual. The fix — thumbstick hysteresis (once a sector is latched, a $>33^\circ$ deflection from its centre is required to switch) plus two-stage confirmation (first trigger press locks the cursor to yellow; second press confirms and records) — was implemented in the subsequent software commit and has been active in all sessions since. Sessions 1 and 2 are the only contaminated baseline records in this dataset.